

02 Temporal, event gate, boundary attention và motion

Query được học mới; audio điều chỉnh query theo từng mẫu. Motion head là một nhánh riêng.



Qn @ Hn.T gives [B,3,8]. Hn @ Qn.T would give [B,8,3] and needs transposing.
Pooling uses H, not normalized Hn. Event gate supplies a shared prior to all 3 queries.
Motion head is computed in forward, but its score does not feed the class prediction.

INITIALIZATION

Checkpoint-loaded: audio frontend/CNN6/classifier; video EfficientNet/classifier.

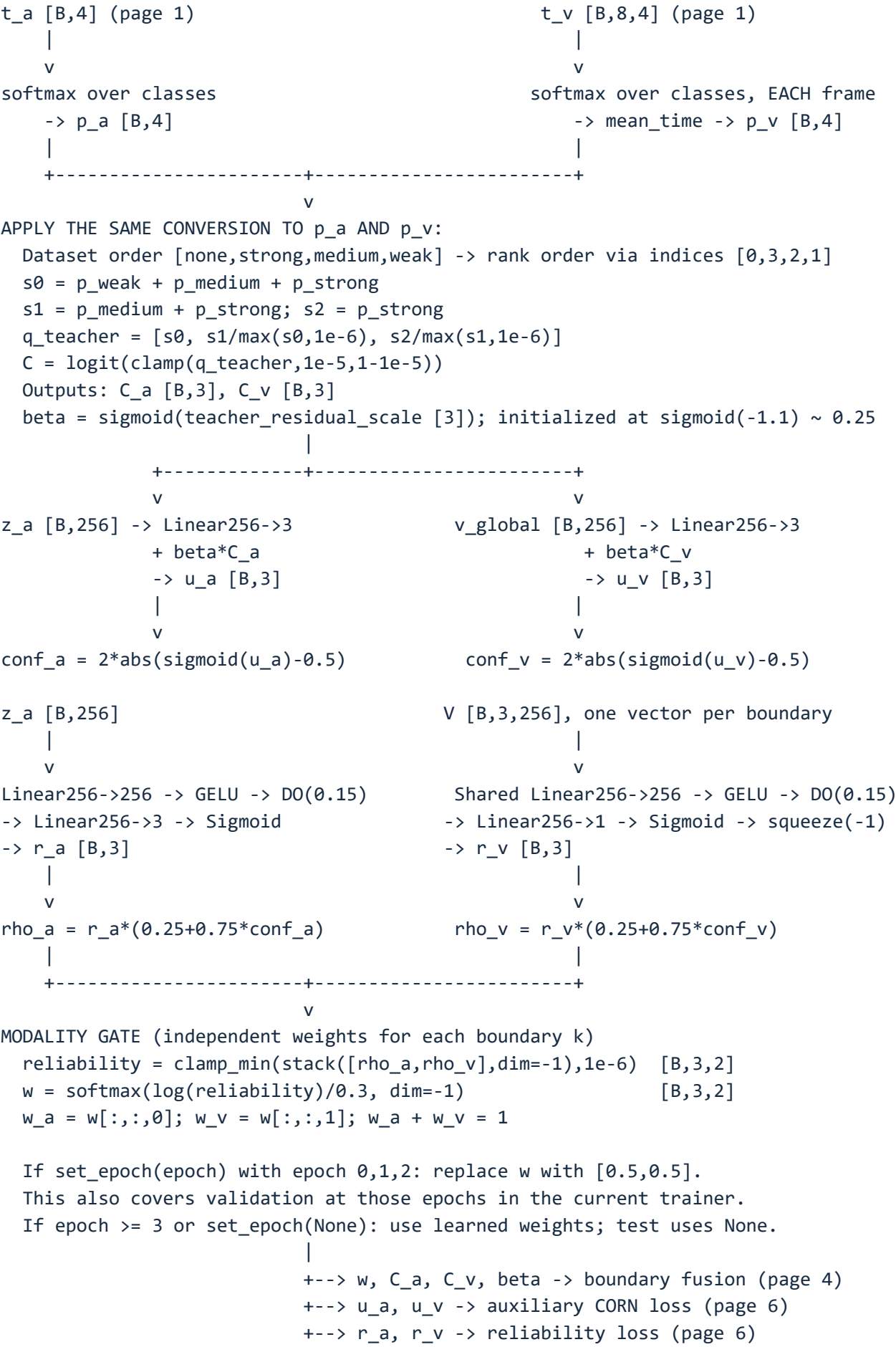
Newly initialized: projections, temporal encoder, event gate, queries, fusion and heads.

boundary_queries: truncated normal std=0.02; learned jointly from supervised losses.

Both original encoders/classifiers continue fine-tuning; STFT/Mel weights stay frozen.

03 Teacher conversion, auxiliary heads và reliability

Teacher là classifier gốc được tái sử dụng. Audio heads nhận z_a; video heads có hai input khác nhau.

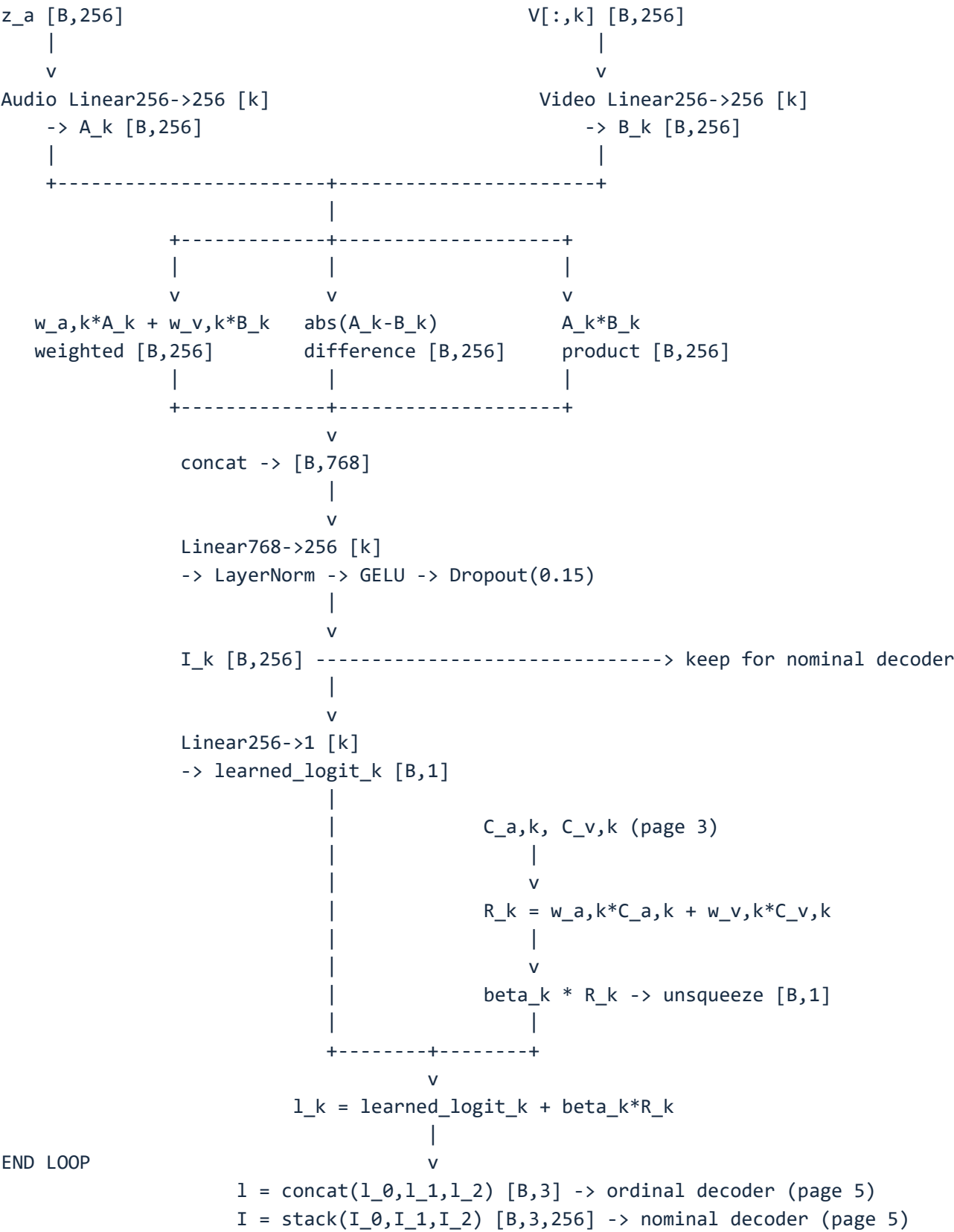


Teacher probabilities/logits are NOT detached in prediction paths.
Preservation loss uses mean_time(t_v) LOGITS, unlike conversion's mean of probabilities.

04 Boundary-specific fusion

Ba bộ projection, interaction MLP và classifier riêng; cùng một audio z_a đi vào cả ba.

FOR k IN {0,1,2}:

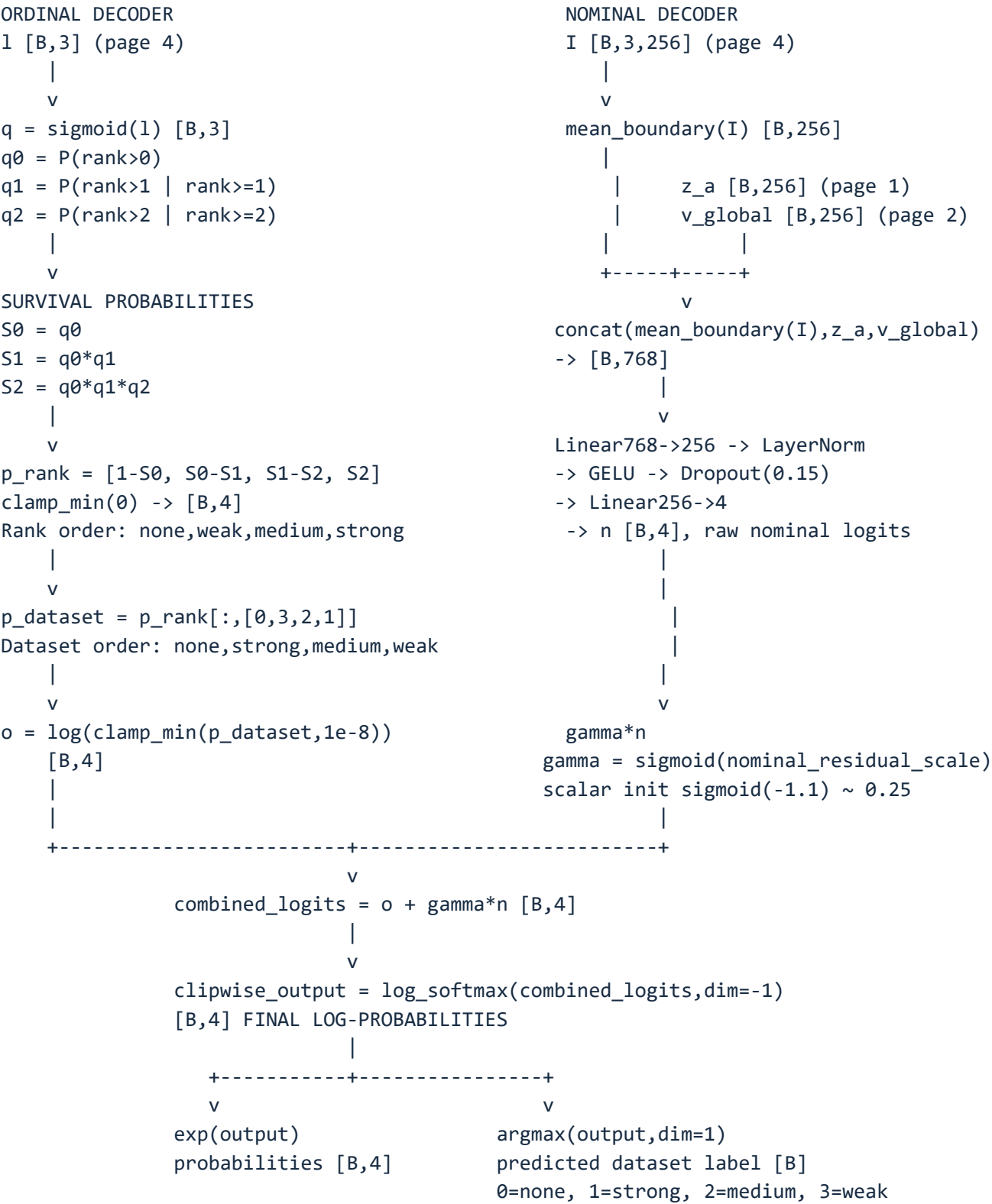


Inputs w, C_a, C_v and beta are the outputs of page 3.
The SAME beta_k is used in audio auxiliary, video auxiliary and fused teacher residual.
Difference/product are not multiplied by modality gate weights.
I_k is captured before its classifier and before teacher-logit addition.

- ORDINAL TASKS (semantic meaning assigned by label-derived loss)
- k=0: P(rank>0) : none vs weak/medium/strong
 - k=1: P(rank>1 | rank>=1) : weak vs medium/strong, when active
 - k=2: P(rank>2 | rank>=2) : medium vs strong, when active

05 Dual decoder và output cuối

Nominal decoder còn nhận trực tiếp z_a và v_global. clipwise_output là log-probabilities.



06 Nhãn, loss và đường học

Boundary học từ nhãn thật qua backpropagation; feature và teacher residual tận dụng model mẹ.

```
Dataset target y [B]: none=0, strong=1, medium=2, weak=3
|
v
Ordinal rank r = [0,3,2,1][y]      (none=0, weak=1, medium=2, strong=3)
|
+--> for k=0,1,2: target t_k = 1[r>k]; active mask M_k = 1[r>=k]
|     CORN(x,r) = mean(BCEWithLogits(x,t) over all active batch-boundary entries)
|
+--> normalized rank r/3 -> motion target

FORWARD OUTPUT / TARGETS                                LOSS TERM
-----
l, r -----> L_fused = CORN(l,r)

u_a, u_v, r -----> L_aux = CORN(u_a,r) + CORN(u_v,r)

u_a, t -> exp(-BCE(u_a,t)).detach() --+
                                     +-----> SL1_a = SmoothL1(r_a,detached_target_a)
r_a -----+

u_v, t -> exp(-BCE(u_v,t)).detach() --+
                                     +-----> SL1_v = SmoothL1(r_v,detached_target_v)
r_v -----+

                                     L_rel = 0.5*(masked_mean(SL1_a,M)
                                     +masked_mean(SL1_v,M))

clipwise_output, y -----> L_cat = NLL(clipwise_output,y)

motion_score, r/3 -----> L_motion = SmoothL1(motion_score,r/3)

n, y -----> L_nom = CrossEntropy(n,y)

t_a, mean_time(t_v), y -----> L_pres = 0.5*(CE(t_a,y)+CE(mean_time(t_v),y))

                                     |
                                     v

TOTAL = L_fused + 0.3*L_aux + 0.1*L_rel + 1.0*L_cat
       + 0.1*L_motion + 0.5*L_nom + 0.3*L_pres

|
v
backward() -> Adam step
+--> original encoder branches + original classifiers: lr=1e-5
+--> new fusion/projections/Transformer/queries/heads: lr=1e-4
```

Boundary queries/event gate/attention learn indirectly through downstream supervised losses. No separate frame-event label, attention target, or copied boundary query from a parent model. Raw r_a/r_v are supervised by reliability loss; confidence-modulated rho_a/rho_v feed the gate. Teacher preservation uses real labels, not a frozen-teacher distillation target.

CURRENT TRAINING SETTINGS
120 epochs maximum; batch_size=32; Adam; early stopping patience=20.
Best checkpoint selected by validation accuracy.
Frozen: STFT and Mel filter weights. Encoders and their original classifiers fine-tune.